DO NOW — RECALL

Lesson 072 · Algorithm Design: From Specification to St

Name: _____ Class: _____ Date: _____

1. What is a loop invariant?

2. What is a postcondition?

3. How is an input specification useful?

4. C# array a=[3,5,7]: set a[1]=a[0]+a[2]. Give final array and sum.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Property true after processing each step.

2. Model answer

Expected outcome after valid execution.

3. Model answer

Defines valid types, conditions and outputs.

4. Model answer

[3,10,7], sum20.

By the end, you can…

• Read the problem to identify exact input type, required output, boundaries and invalid cases.

• Write an algorithm with an invariant: after each iteration, count is the number of qualifying values examined so far.

• Choose meaningful tests around the 40 boundary and compare output against an independently calculated result.

Vocabulary — know these terms

algorithm — Finite unambiguous sequence of steps.

invariant — Property preserved across loop iterations.

precondition — Required input assumption.

postcondition — Expected condition after executing.

NEW LEARNING

Read, annotate and explain

Observe

Read the problem to identify exact input type, required output, boundaries and invalid cases.

Explain

Write an algorithm with an invariant: after each iteration, count is the number of qualifying values examined so far.

Evaluate

Choose meaningful tests around the 40 boundary and compare output against an independently calculated result.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Design a function counting even numbers in [1,2,6,7].

Explain

Using modulo2==0 identifies even values; the list has 2 and 6.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Trace CountPasses for [40,41,39,100]. Show count after each item. [6]

Question 2

Write C# method that counts negative integers in an array, and returns0 for empty. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain a loop invariant, time complexity and edge tests for CountPasses. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

An algorithm can be evaluated without considering its input requirements.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.